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Student Handout

Name: A Proletariat **Date:** 03/25/2026 **Class:** Period 6 Engineering Systems

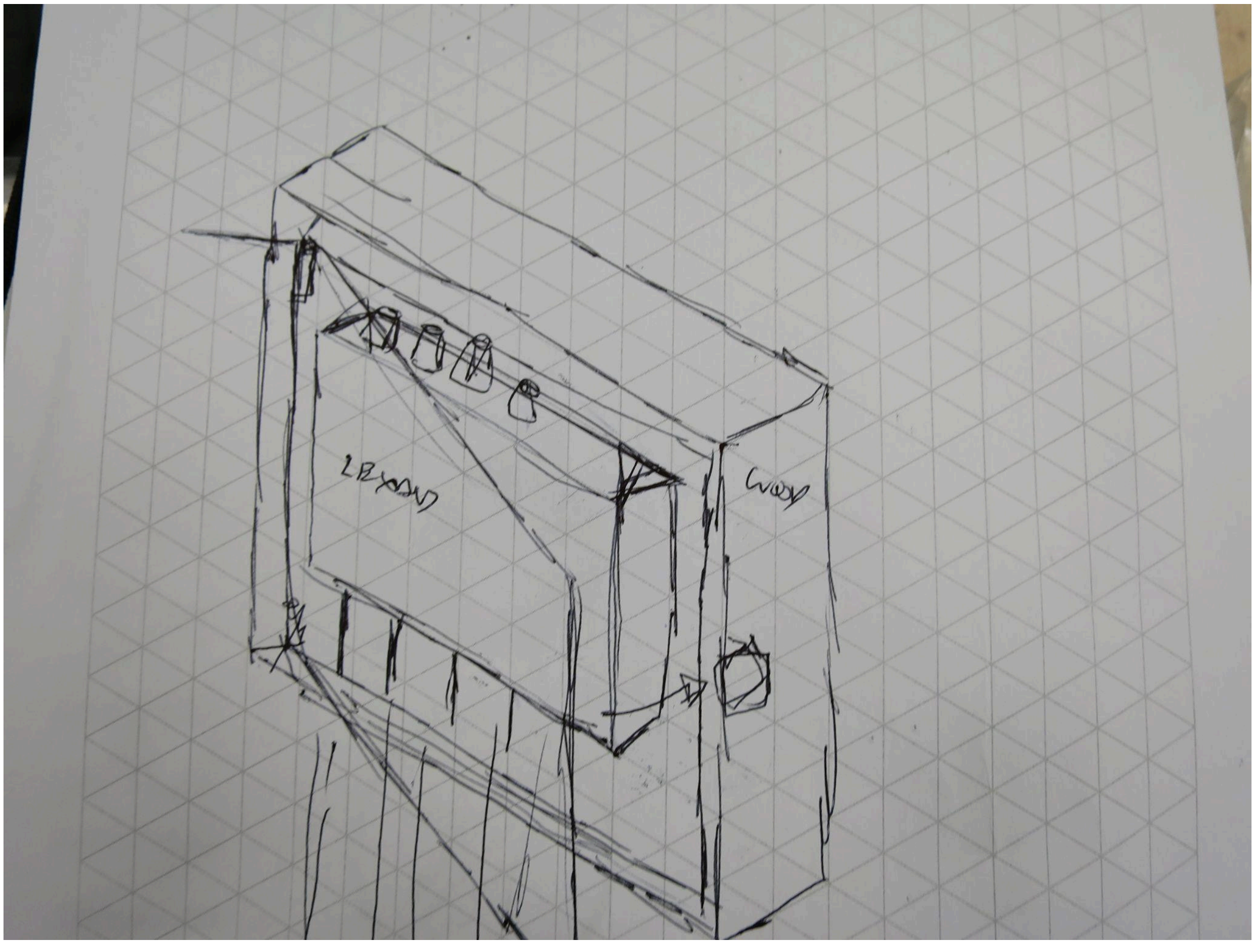
MyDesign Portfolio

Element D

Element D: Design Concept Generation, Analysis, and Selection

Box Protection + Cable Organizer:

Date: 03/25/2026
Description: Utilizing a box with holes to route cables out of. Directly secured to the wall using double sided tape/screws (with school permission) and a 3D Printed Frame. Should include a sliding, or opening door to gain easy access to wires. Should route the wires correctly as well.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



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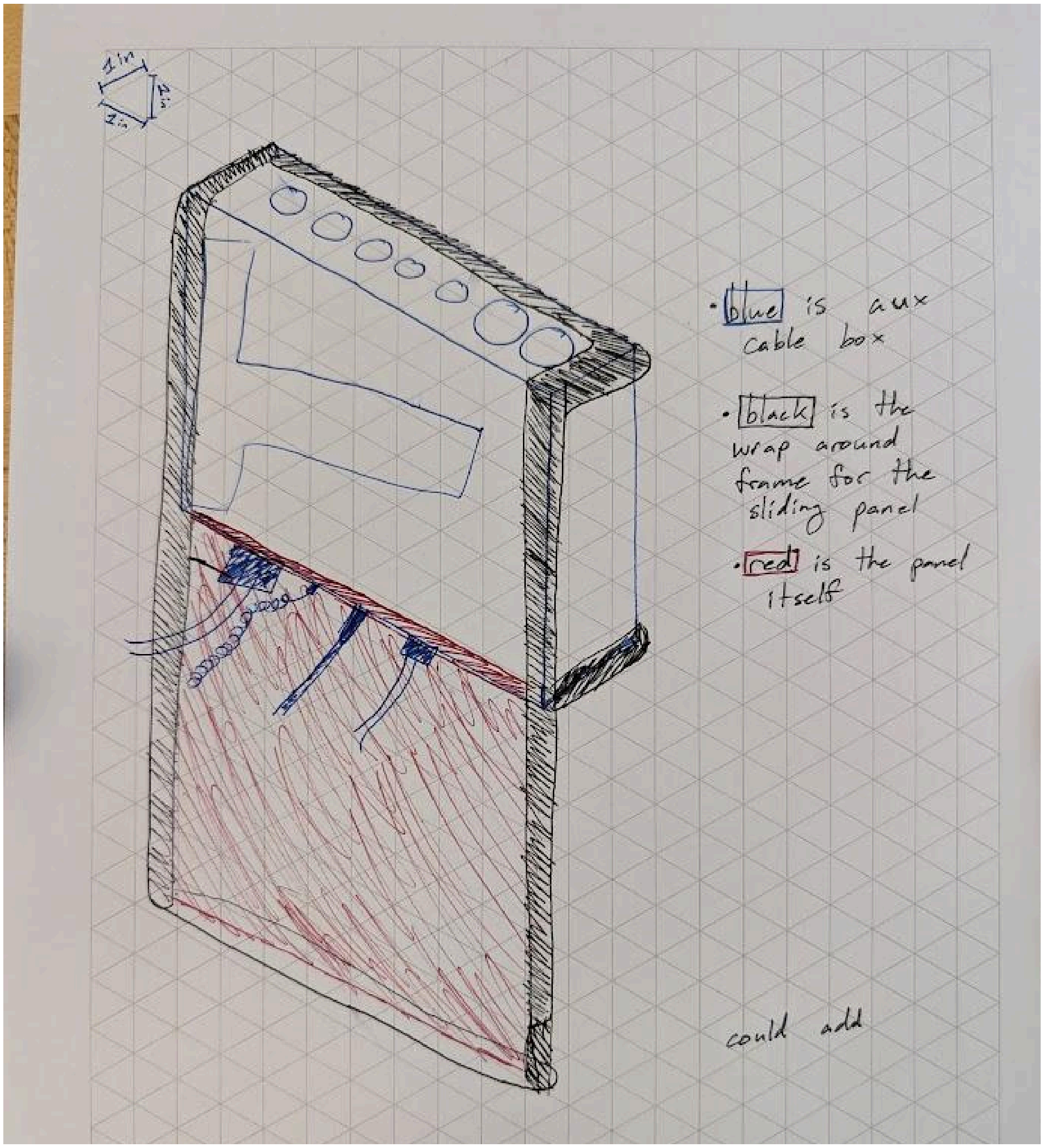
Sliding Panel Cable Protection:

Date: 03/25/2026
Description: Mounting a sliding panel to the front of the box to hide the wires and block impacts. Would be mounted with one 3D printed part around the box and another sliding up-down along the front of the box. Could add a tray at the base for added structure and to hold up the AC adapter and other wires. Not intended to wire organize.
Drawing, including distinguishing information such as scale, dimensions, and materials:



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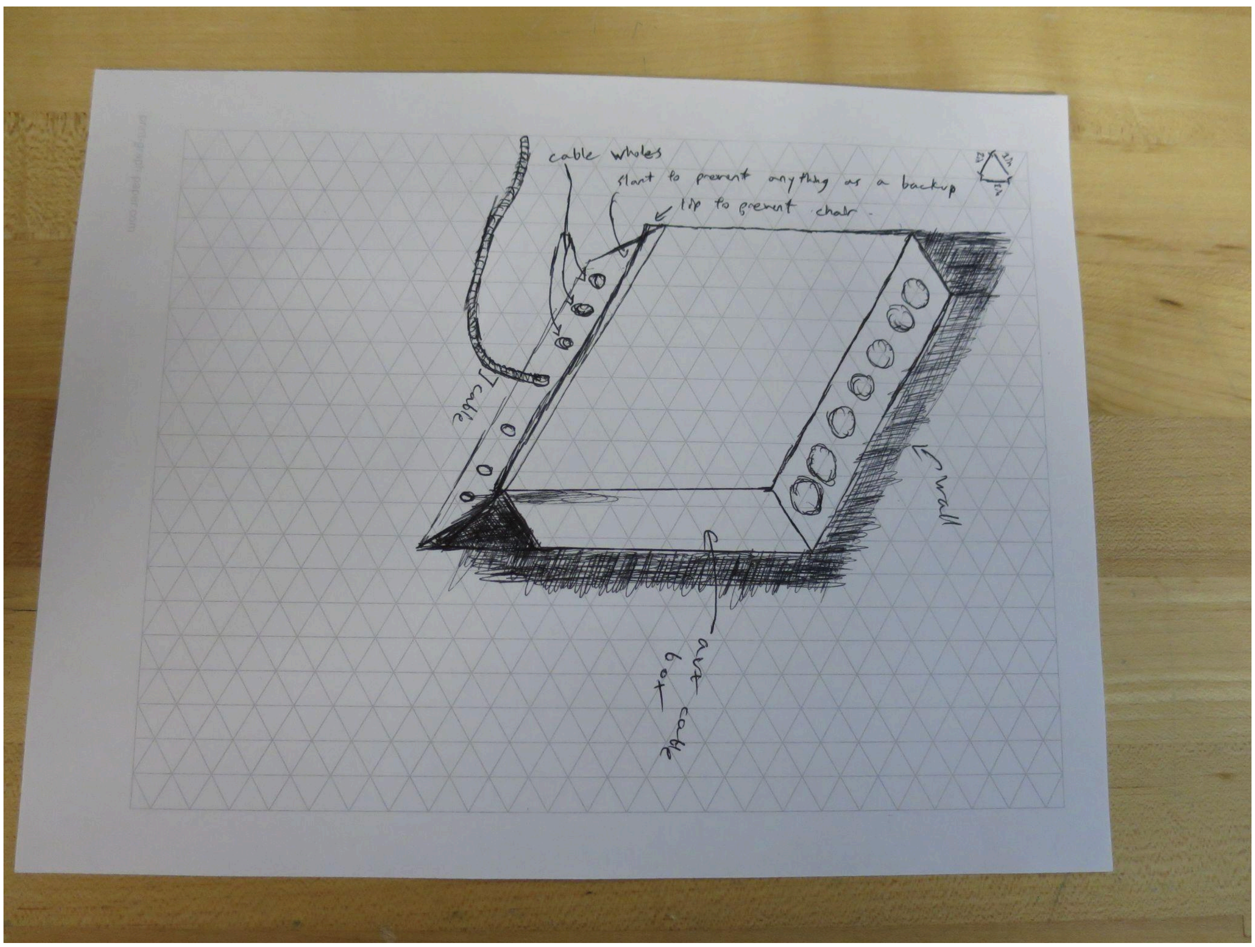
Angled Bottom Mount:

Date: 3/25/2026
Description: Create a small lip fastened to the wall beneath the AUX cable box that prevents the chair from sliding back under the box. The lip would also have a number of holes on the inside going up-down to segregate all of the wires coming out of the AUX cable box and allows for easier control over their location.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



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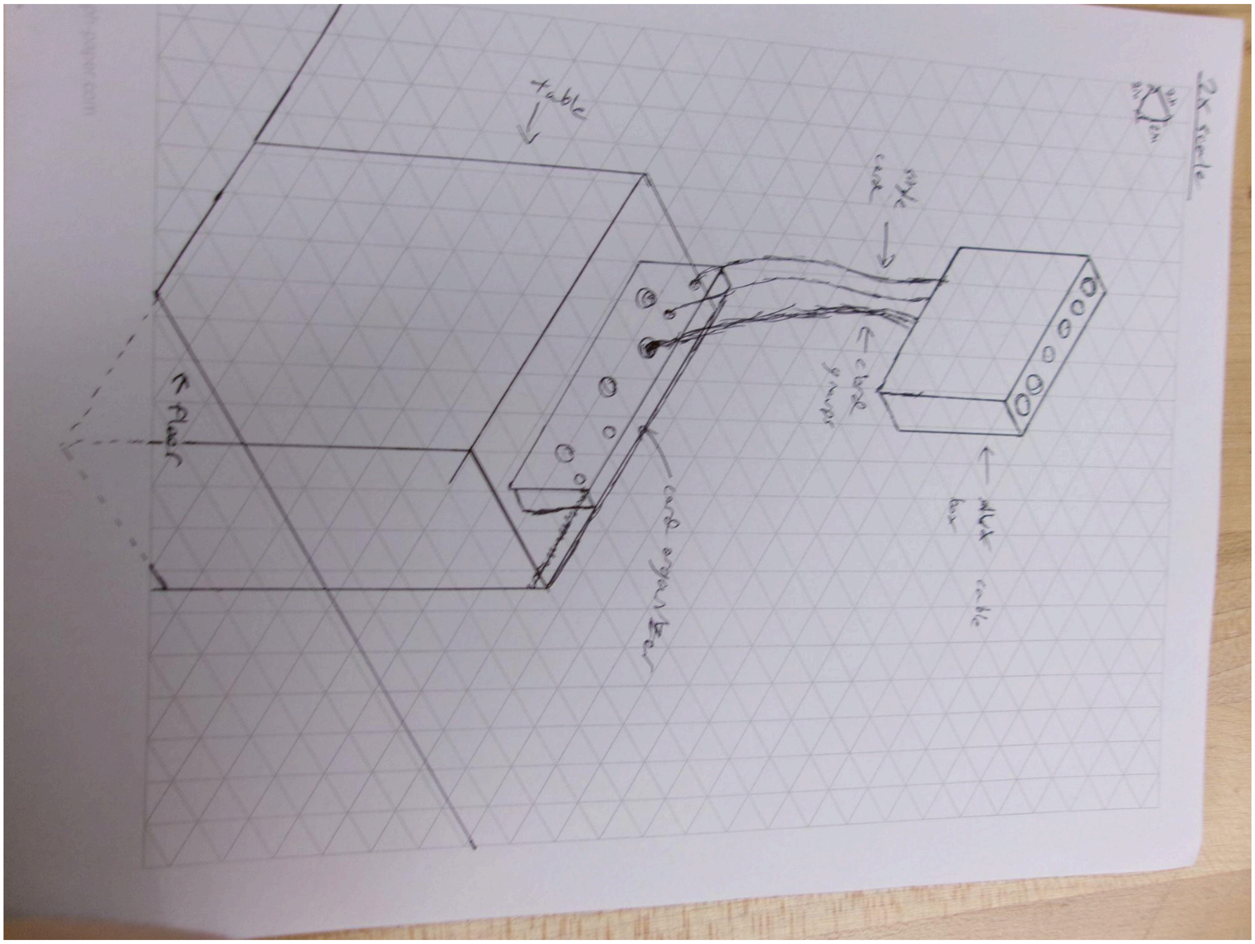
Cable Table Blockade

Date: 3/25/2026
Description: A cable organizer that focuses primarily on organizing the cables on a table stand that sits underneath the AUX cable box to prevent the chair from sliding between the floor and the cable box. The cable organization box will have an open top with many holes that can collect all the cables and then be separated to wherever they need.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



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Design Idea Comparison and Selection *(D1, D2)*

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

Des. Reqs	weight	Option 1	Option 2	Option 3	Option 4
Stop Chair from Wedge.	0.30	0.30 (Solid, attached blockade)	0.30 (Solid, attached blockade)	0.30 (Solid, attached blockade)	0.30 (Solid, grounded object)
Direct Cable Protection	0.30	0.20 (Decent routing)	0.25 (Strong routing)	0.15 (Not large enough)	0.30 (Solid object)
Security + Ease of Access	0.15	0.10 (Could have some problems with access)	0.15 (Can open/shut easily)	0.15 (Not obstructing anything)	0.10 (Table could be cumbersome)
Conceal Solution	0.10	0.00 (Quite visible)	0.05 (Blends in)	0.05 (Somewhat smaller space)	0.00 (Very visible)
Organize Cables	0.15	0.15 (Enough space)	0.15 (Enough space)	0.10 (Can run into problems routing)	0.15 (Enough space)
score	1.00	0.75	0.90	0.75	0.75

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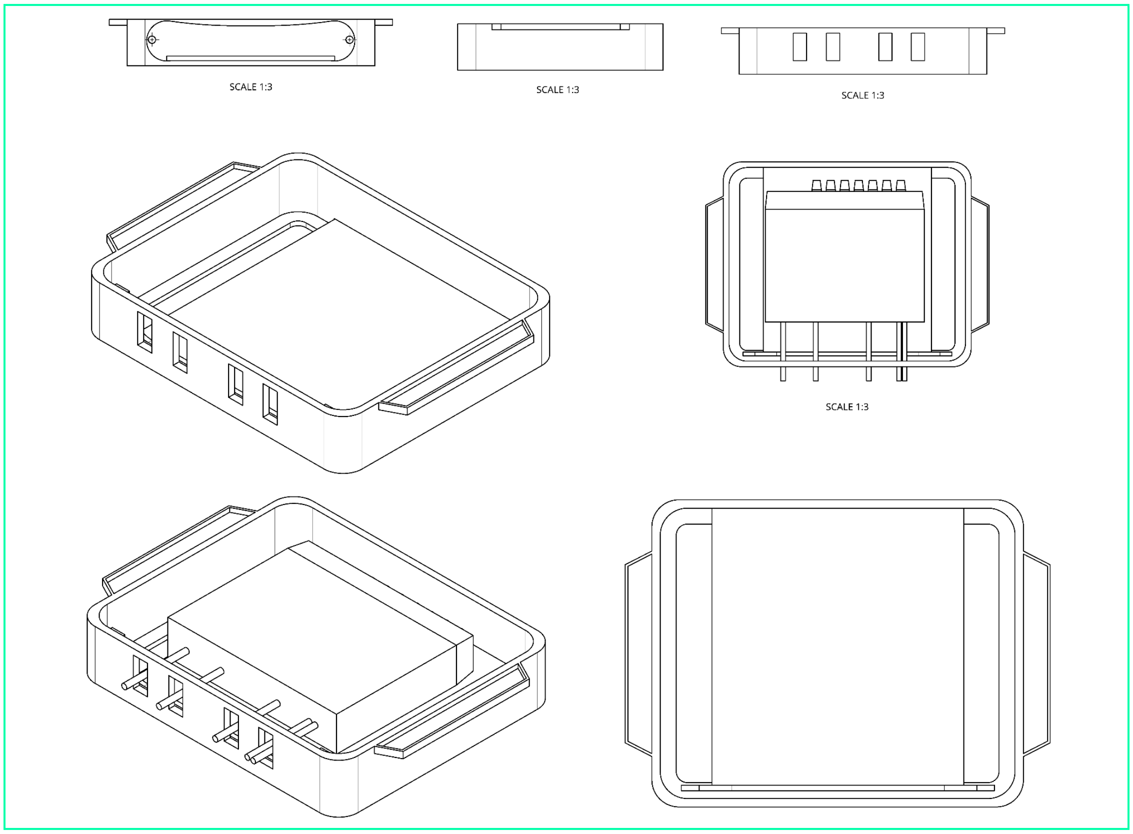
Design Blueprint *(D4)*

Date: 4/20/2026

Sketch(es):



Manufacturing Methods:

1. CNC machine out each part (2 of the Curved Sides and the top and bottom panels.)
    a. Drill out holes for dowels on each piece
    b. Route out the cutouts for the wires
2. 3D Print the panel sides (not the front panel) + press-fit caps for magnets in the box
    a. Insert magnets into the magnet slots during the print to keep affixed in place
    b. print handle onto the wider side piece
3. Drill out minimum distance for magnets in the box top (might be all the way through) and glue in 3D printed caps to keep the magnets in the wood
4. Spray paint all pieces white before assembly
5. Use dowels + wood glue to fit the box parts together; let dry completely
6. Glue panel together through the box
7. Sand off excess glue and touch up paint

Supply List and Costs:

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Item	**Cost each**	**# needed**	**Total**
1/4" Birch Wood Sheets	$10.00	4	$40.00
Neodymium Magnets	$0.70	12	$8.40
Wood dowels, 1/8"	$0.10	4	$0.40
Wood glue	$8.00/bottle	¼ bottle	$2.00
White PETG	$19.00/spool	¼ spool	$4.75
white (spray) paint	$9.50/can	1 canister	$9.50
Total Cost			$65.05